



MANJUSRI S

+91 9042751605

manjusri.s1605@gmail.com

Coimbatore,Tamil Nadu

PORTFOLIO

[Linkedin](#)

About Me

Final-year Electronics and Communication Engineering student with a strong foundation in Embedded Systems, IoT, Artificial Intelligence, and Machine Learning. Experienced in microcontroller programming, sensor integration, and intelligent monitoring systems, with a keen interest in developing innovative engineering solutions and research-oriented technologies.

Education

B.E-ECE (2023-2027):

Sri Eshwar College of Engineering

CGPA 8.3

HSC (2022-2023):

Barathi Vidya Bavan Higher Secondary School

92%

SSLC (2020-2021):

Vighneshwar Vidhya Mandhir -SSLC

PASS

Skills

- **Programming:** C, C++, Embedded C, Python, ALP
- **Microcontrollers:** Intel 8051, ESP32, STM32
- **Protocols:** UART, I2C, SPI, CAN, MQTT
- **Wireless Technology:** BLE, WiFi, Zigbee
- **Tools:** Keil, STM32CubeIDE, Arduino IDE, Matlab, VS Code, Github, Canva
- **Hardware Integration:** Sensor Integration, Actuators,ADC&DAC,PCB Design
- **Testing Instruments:** Multimeter,oscilloscope,Signal Generator
- **Operating System:** FreeRTOS, Basic Linux

Achievements

- Secured 2nd Position in intercollege event 'Circuit Digest' held as a part of of KRISHNECS 2K24 – A National Level Technical Symposium at Sri Krishna College of Engineering and Technology.

Research Publication:

- A Scalable and Intelligent Approach to Optimizing EV Charging Infrastructure-IEEE-ICEESI 2026

Certifications

Mastering Data Structures & Algorithms using C and C++

[Udemy](#) | 2024

C Programming and Assembly Language

[NPTEL](#) | 2025

Full-Stack Development

[Simplilearn](#) | 2025

Introduction to Cyber Security

[Simplilearn](#) | 2026

SQL(Basic) – Standard

[Skillrack](#) | 2024

Internship

• ARTIWIZ TECHNOLOGIES - EMBEDDED SYSTEMS & IOT (2025)

Developed and optimized real-time applications on Intel 8051 and STM32F407VGT6 using Bare Metal C, HAL, and FreeRTOS.

• CENTRAL SCIENTIFIC INSTRUMENTS ORGANISATION (CSIR- CSIO) (2025)

Supported research and development activities in embedded systems and IoT by applying core electronics principles in real-time sensor integration and hardware deployment.

Projects

• AI-Enabled Agro-Climate Intelligence & Adaptive Farm Decision Support System:

Developed an AI-powered agro-climate forecasting system using TFT to predict crop yield and environmental parameters. Built a battery-operated IoT system with sensor integration and real-time dashboard visualization.

Tech Stack: Machine Learning , Time-Series Forecasting , Temporal Fusion Transformer (TFT), Data Analysis , IoT Sensors

• Predictive Analytics and AI-Optimized Battery Management System for Electric Vehicles:

Engineered an IoT-enabled Battery Management System using ESP32 with real-time electrical and thermal monitoring. Integrated machine learning-based predictive analytics for battery health assessment and performance optimization.

Tech Stack: ESP32 , Current and Temperature sensors , ML (Random Forest) , SOC/SOH Estimation , Cloud (MQTT)

Technical Focus

- Intelligent Embedded Systems
- Edge AI & TinyML
- Sustainable Energy Technologies